Jaehyeon Moon

Yonsei University

Combined M.S. & Ph.D. student

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Research Interests

- Model quantization
- KV cache compression

Education

Mar. 2020 – Present
Yonsei University
School of Electrical and Electronic Engineering
Seoul, Korea
Advisor: Bumsub Ham
Combined M.S. & Ph.D. student

Mar. 2016 – Feb. 2020
Yonsei University
School of Electrical and Electronic Engineering
Seoul, Korea
B.S. in Electrical and Electronic Engineering
Magna Cum Laude (GPA: 4.06 / 4.30, Top 3%)

Publications

First-author

- **Jaehyeon Moon**, Hyeonwoo Cho, Jeongwoo Choi, Yura Seo, and Bumsub Ham, "KVRecon: KV Cache Merging via Reconstruction Error Minimization", ICLR, 2027 (under review).
- **Jaehyeon Moon** and Bumsub Ham, "Shift-and-Sum Quantization for Visual Autoregressive Models", ICLR, 2026.
- Dohyung Kim*, **Jaehyeon Moon***, Junghyup Lee, Geon Lee, Jeimin Jeon, and Bumsub Ham, "Token-Based Dynamic Bit-Width Assignment for ViT Quantization", Pattern Recognition, 2026. *Equal contribution.
- **Jaehyeon Moon**, Dohyung Kim, Junyong Cheon, and Bumsub Ham, "Instance-Aware Group Quantization for Vision Transformers", CVPR, 2024.

Co-author

- Yura Seo, Hyeonwoo Cho, Jeongwoo Choi, **Jaehyeon Moon**, and Bumsub Ham, "Compensating Token Merging for Efficient Diffusion Transformers", ICLR, 2027 (under review).
- Seunghoon Lee*, Jeongwoo Choi*, Byunggwan Son, **Jaehyeon Moon**, Jeimin Jeon, and Bumsub Ham, "AccuQuant: Simulating Multiple Denoising Steps for Quantizing Diffusion Models", NeurIPS, 2025. *Equal contribution.

- Dohyung Kim, Junghyup Lee, Jeimin Jeon, **Jaehyeon Moon**, and Bumsub Ham, "Toward INT4 Fixed-Point Training via Exploring Quantization Error for Gradients", ECCV, 2024.

Experiences

Aug. 2023 – Present
Articron
Research Engineer | TIPS Project

- Developed a hardware-aware PTQ framework for CNNs, with optimized LUT approximations for nonlinear activation functions.
- Developing an LLM quantization framework supporting various PTQ methods and numerical formats (INT, FP, MXFP), targeting FPGA-based inference.

Patents

Granted

- Bumsub Ham, **Jaehyeon Moon**, Dohyung Kim, and Junyong Cheon, "Quantization Apparatus and Method for Artificial Neural Networks and Image Processing Device Having the Same", U.S. Patent No. 12,684,127, granted Jul. 2026.
- Bumsub Ham, **Jaehyeon Moon**, Dohyung Kim, and Junyong Cheon, "Quantization Apparatus and Method for Artificial Neural Networks and Image Processing Device Having the Same", Korean Patent No. 10-3028313, granted Sep. 2026.

Applications

- Bumsub Ham and **Jaehyeon Moon**, "Quantization Method and Apparatus for Visual Autoregressive Model", U.S. Patent Application No. 19/548,809, Apr. 2026.
- Bumsub Ham and **Jaehyeon Moon**, "Quantization Method and Apparatus for Visual Autoregressive Model", Korean Patent Application No. 10-2026-0032167, Feb. 2026.
- Bumsub Ham, Dohyung Kim, Junghyup Lee, **Jaehyeon Moon**, Geon Lee, and Jeimin Jeon, "Token-Based Quantization Apparatus and Method for Vision Transformers", Korean Patent Application No. 10-2024-0137421, Oct. 2024.
- Bumsub Ham, Dohyung Kim, Junghyup Lee, Jeimin Jeon, and **Jaehyeon Moon**, "Quantization Apparatus and Method for Artificial Neural Networks", Korean Patent Application No. 10-2023-0116857, Sep. 2023.

Awards & Scholarships

- National Science & Technology Scholarships, Ministry of Science and ICT, Republic of Korea (Sep. 2018 – Feb. 2020)